MIL-B-30510B
18 August 1983
SUPERSEDING
MIL-B-30510A
36 March 1970

MILITARY SPECIFICATION

BLOCK AND TACKLE SET: LINEMAN'S

WITH SNUBBING HOOK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a block and tackle set with snubbing hook, and two cable grips when specified herein.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3940

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SPECIFICATIONS

FEDERAL

T-R-605	- Rope, Manila and Sisal.
RR-G-691	- Grip, Cable Jaw.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 123	- Zinc (Hot-Galvanized) Coatings on Products Fabricated from Rolled, Pressed, and Forged Steel Shapes, Plates, Bars, and Strip.
ASTM E 4	- Load Verification of Testing Machines.

(Application for copies should be addressed to the American National Standards Institute, Inc. 1430 Broadway, New York, NY 10018).

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Description. The block and tackle set shall be as shown in figure 1 (see 6.4) and as specified herein. It shall consist of two double-sheave blocks reeved with a single length of rope and an anchor hook for attachments to the snap hook of either block.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Material. The material shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the block and tackle set to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product (see 6.5).

3.3.1 Material deterioration and control. The block and tackle set shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environment to which the item may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available upon request to the contracting officer or designated representative.

3.4 Anchor hook. The anchor hook shall be regular style 5/8-inch diameter round eye, and the eye shall be in the same plane as the hook body.

3.5 Rope. The rope shall be mildew resistant, regular lay, three strand, or medium lay, four strand, 3/8-inch diameter manila conforming to T-R-605. Each block and tackle set shall have 25 feet of rope, excluding terminations. The rope shall be spliced into the eye of the bracket (see figure 1).

3.5.1 Splices. A lock tuck eye shall be formed by bending the end of the rope upon itself to form an eye. End of the rope shall be eye spliced in accordance with an approved manual.

3.6 Blocks. Each block shall consist of two shell plates, one sheave separator, two sheaves, two sheave bearings, one snaphook with integral spacer, one sheave pin and two rivets. One block shall also be equipped with a pair of plain spacers and the other block with becket and integral spacer. One shell plate shall be designed to form a snubbing hook for the rope.

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3.6.1 Shell plates. Shell plates shall be forged steel, galvanized in accordance with ASTM A 123.

3.6.2 Sheave separators. Sheave separators shall be forged steel or malleable iron, galvanized in accordance with ASTM A 123.

3.6.3 Sheaves. Sheaves shall be not less than 1-1/2-inch rim diameter and grooved for 3/8-inch diameter rope. Sheaves shall be steel or malleable iron, galvanized in accordance with ASTM A 123.

3.6.4 Bushings. Each sheave shall be furnished with a one-piece, self-lubricating high grade bronze bushing which shall be pressed into the sheave with sufficient tightness to prevent slippage at loads of 4 times the safe working load (swl) (see 3.7.1). The bushing shall be equal in length to sheave thickness at the hub and shall have a thickness of not less than 3/16 inch.

3.6.5 Snaphook and spacer. Snaphook and spacer shall be a single unit, forged steel, galvanized in accordance with ASTM A 123, and designed as shown in figure 1 and as specified herein. The hook shall have a clear opening of not less than 3/4 inch and shall be recessed to provide a seat for the safety latch (see 3.6.10).

3.6.6 Center pin. The center pin shall be designed to transmit the sheave test (see 4.5.2.1) load to the shell plate and serve as an axle for the sheave. Means shall be provided to prevent pin rotation.

3.6.7 Riveted connections. Rivet holes shall be accurately punched or drilled, and burrs that prevent surfaces of connected parts being fitted up shall be removed. Rivets shall be driven to completely fill the holes. Heads shall be concentric with the shank and have full bearing all round. Each type head of the same diameter rivets shall be uniform in size and shape.

3.6.8 Spacer and becket. Spacer and becket shall be a single unit, forged steel, galvanized. Design shall be as shown in figure 1 and as specified herein. Dimensions of the becket shall be compatible with the 3/8-inch diameter rope.

3.6.9 Plain spacer. The plain spacers shall be galvanized steel.

3.6.10 Safety latch and spring. Each snaphook shall be equipped with a spring closing safety latch. The latch shall be galvanized steel and the spring of corrosion-resistant material. Safety latch, spring, and pin shall be installed in such a manner so the assembly operates freely and shall seat and unseat without binding.

3.6.11 Cable grips. When specified (see 6.2), two cable grips shall be provided as part of the block and tackle set. Cable grips shall conform to RR-G-691, type I, style A, size 4.

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3.7 Capabilities.

3.7.1 Safe working load. The swl of the block and tackle set shall be not less than 1200 pounds.

3.7.2 Proof load. When subjected to a proof load twice the swl specified in 3.7.1, the block and tackle set shall show no evidence of deformation, distortion, breakage, or permanent set.

3.8 Weight. The weight of the block and tackle set shall be not more than 4-1/2 pounds.

3.9 Identification marking. Each block shall be permanently and legibly marked with the manufacturer's name or trademark of such known character that the source of manufacture of the block may be readily determined. In addition, the safe working load shall be permanently marked or stamped on the block. Each block shall be identified in accordance with MIL-STD-130.

3.10 Workmanship. The block and tackle set shall be free from defects which could affect its serviceability, uniformity of components, and smooth functioning.

3.10.1 Forgings. Forgings shall be uniform quality and shape and free from flash, scale, cracks, hard spots, burrs, sharp edges or distortions.

3.10.2 Fastening devices. Screws, pins, bolts, and rivets shall be tight. Parts subject to removal or adjustment shall not be swaged, peened, staked, or otherwise deformed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article block and tackle set shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article block and tackle set shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Lot. A lot shall consist of all block and tackle sets which have been manufactured under the same conditions and offered for delivery at the same time.

4.4.2 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, inspection level S-2.

4.4.3 Examination. Samples selected in accordance with 4.4.2 shall be examined as specified in 4.5.1. AQL shall be 0.4 percent defective for major defects, and 1.0 percent defective for minor defects.

4.4.4 Tests. Samples selected in accordance with 4.4.2 shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection of the lot.

4.5 Inspection procedure.

4.5.1 Examination. The block and tackle set shall be examined as specified herein for the following defects:

Major

101. Material not as specified or the materials are not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environment.

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102. Finish plan not adequate or the contractor does not have documentation available for identification of material, material finishes or treatments.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated for each other.
104. Parts or components missing.
105. Rope serviceability affected by cut, chafing, or any other damage.
106. Removable parts not as specified.
107. Sheave bushing not self lubricated.
108. Center pin rotates.
109. Dimensions of becket not compatible with size of rope.
110. Safety latch and spring inoperative.
111. Weight not as specified.
112. Workmanship not as specified.

Minor

201. Extraneous material in rope.
202. Identification marking missing, incomplete or illegible.

4.5.2 Tests.

4.5.2.1 Proof. Block and tackle sets shall be tested in a testing machine verified in accordance with ASTM E 4. The proof load specified in 3.7.2 shall be applied to pass through the sheaves, sheave pin, and fittings. A curved steel bar or wire rope suitable to the block being tested shall be applied around the sheaves not less than 180 degrees and used to connect the block to the fixed head of the testing machine. The hook of the block shall be attached to the movable head of the machine by a suitable link or shackle. Evidence of deformation, distortion, breakage or permanent set shall constitute failure of this test.

4.5.2.2 Operation. The block and tackle shall be suspended from a fixed overhead point. A 1200 pound load shall be manually raised and lowered not less than 5 feet and not less than 3 times. Upon completion of these tests the rope shall be snubbed on the snubbing hook to determine compliance with the following:

a. Blocks operate freely.

b. Snubbing hook holds rope under swl condition.

c. Splice remains secure under swl condition.

Failure to conform to any of the above shall constitute failure of this test.

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4.5.2.3 Bushing. Operate the block and tackle as described in 4.5.2.2, except that a 4800 pound load shall be used. Evidence of slippage of the bushings shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

113.	Containers not as specified for level A or B.	Each incorrect container shall constitute one defect.
114.	Cushioning not provided to prevent movement of the block and tackle sets within the container for level A.	
115.	Box closure and sealing not as specified for level A.	
116.	Preservation and packing not in accordance with the referenced document as specified for commercial.	
117.	Quantities packed exceed the size or weight limitation of the box for level A or B.	
118.	Marking illegible, incomplete, incorrect, or missing for level A, B, or commercial.	

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. Each block and tackle set shall be placed in a close-fitting box conforming to PPP-B-636, W5c, or W6c, style optional. The contents shall be cushioned and secured to prevent damage by free movement within the box. Box closure and sealing shall be as specified for method V in accordance with the appendix to the box specification.

5.1.2 Commercial. Each block and tackle set shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

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5.2.1 Level A. The block and tackle sets preserved as specified in 5.1 shall be packed in close fitting boxes conforming to PPP-B-601, overseas type, style optional in quantities not to exceed the weight limitations of the box specification. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.2.2 Level B. The block and tackle sets preserved as specified in 5.1 shall be packed in close fitting boxes conforming to PPP-B-636, V3c, V11c, V13c, or V15c, or in boxes conforming to PPP-B-640, class 2, style optional, in quantities not to exceed the size and weight limitations of the applicable box specification. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.3 Commercial. The block and tackle sets preserved as specified in 5.1 shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. Block and tackle sets are intended for use in light pole line construction and maintenance.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. When a first article is required for inspection and approval and number of units required (see 3.2).

c. When cable grips are required (see 3.6.11).

d. Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 First article. When a first article inspection is required, the item will be tested and should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangments for examinations, test and approval of the document's first article.

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6.4 Information figure. Figure 1 shows a type of block and tackle set which has been found acceptable; however, the figure is included for illustration only and is not intended to preclude the furnishing of another block and tackle set which conforms to this specification.

6.5 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

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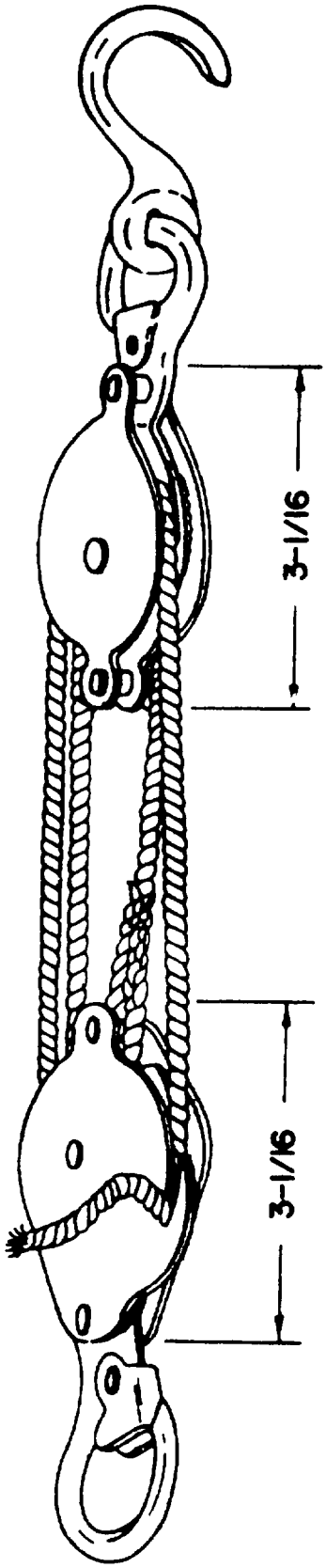
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FIGURE 1. Block and tackle set: lineman's, with snubbing hook.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-30510B	Block and Tackle Set: Lineman's with Snubbing Hook

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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